

Nan Headwaters Community Learning Center

Chumchon Ton Nam Nan (ChTN) · Nan Agri-Nature Center · Tha Wang Pha, Nan

Soil, water, forest and people — restoring a headwater landscape through natural agriculture

Background

Nan Province is more than 80% mountainous and only about 10% of its land is arable. Most of that farmland has been given over to monocultures of cash crops such as rubber and maize, setting off a vicious circle: deforestation brings soil erosion and landslides, agrochemicals wash into the waterways, and the cycle begins again. The stakes reach far beyond Nan. Its river supplies about 47% of the flow of the Chao Phraya, the “River of Kings” that irrigates central Thailand’s rice bowl, and as upstream forests disappeared, floods and droughts followed downstream.

The Nan Headwaters Community learning center, also known as the Nan Agri-Nature Center, was founded in Tha Wang Pha District in 2013 in response. Its guiding idea is that Nan’s problems are national, not merely local, and that even a small demonstration site can spread practices outward. Drawing on King Rama IX’s philosophy of soil, water, forest and people and on the Sufficiency Economy Philosophy, it shows how degraded farmland can become a diverse, water-retaining landscape that provides food security, income from surplus produce and richer biodiversity.





Origins of the Learning Center

Mrs Kul's path back to Nan

The center's founder, Mrs Kul Panyawong, has worked in the environmental field for some twenty years. A former mover and shaker in Bangkok's advertising industry, she spent five years at J. Walter Thompson, then worked in community relations at CHR Community Service. A keen cyclist, she founded a mountain-biking company, ran conservation tourism, and produced travel books, magazines and television programs.

She then spent six years at the Thai Forest Conservation Foundation in Chiang Mai, a well-resourced organization funded by the electricity company EGCO. Over those years she felt increasingly drawn back to her home province, where forest cover was shrinking fast. Returning seemed difficult, though: other organizations were already active in Nan, and she had no local network of her own.

Learning from Ajarn Yak

The turning point came when she met Ajarn Yak (Dr. Wiwat Salyakamthorn), chairman of the Agri-Nature Foundation. He argued that Nan's problems were national in scale, and that a small demonstration point spreading ideas outward could help address them. From him she learned how to put the Khok Nong Na model into practice. She then replicated the model on her own land, refining her skills before passing them on through training at the center she founded.

Overcoming obstacles

Her first attempt to change farming in Nan was to raise funds to pay villagers to farm without chemicals. It did not work: money, she quickly learned, is not the only factor in play. Her new plan was to show people that the system really works.

> ***"Lasting change comes when people actually see you do something and succeed at it. When they see that the Khok Nong Na model works, that's when things start happening."***
>
> *Kul Panyawong, founder of ChTN*

For many farmers, watching their livelihoods improve alongside biodiversity was a revelation. "They see that the land can be a source of food, where they can plant rice, grow vegetables and raise animals," Kul says. "Everything will become more fertile, the soil, water, forests. They will have more food and make more money. Ultimately, their lives will be more stable."

The model reaches the prisons

The same model has since been taught at its smallest scale in prisons, under King Rama X's Khok Nong Na Nam Jai Haeng Khwam Wang ("Hopeful Heartland") project. During COVID-19, a 14-day program was run in 137 prisons nationwide, including in Nan, for inmates facing unemployment on release, most of them from farming families. It began at Klong Prem Prison on a plot of just 12 × 11 meters and covered the Sufficiency Economy Philosophy, natural-agriculture principles, land design and soil improvement.

KHOK NONG NA LAND ALLOCATION

30%	30%	30%	10%
water	forest	rice fields	residence

The Site: Soil, Water and Forest

A landscape designed for learning

From the outset Khun Kul designed ChTN as a training center with multiple learning stations, working with a dean of the Faculty of Architecture at King Mongkut's Institute of Technology Ladkrabang. The layout follows King Rama IX's New Theory and the Khok Nong Na model: khok means "mound", nong "marsh" and na "rice field".

THREE FORESTS, FOUR BENEFITS

The integrated-farming system popularized by King Rama IX combines three kinds of "forest": fast-growing trees such as neem and bamboo; vegetables, fruits and herbs; and commercial timber. Together they bring four benefits: wood to build homes and furniture; food and medicine; raw materials for household products such as firewood, kitchen utensils and soap; and shade to protect fields, animals and buildings.

Healing contaminated land

The land had previously been rubber and longan orchards, heavily treated with chemicals, and the 14-rai entrance area was a pesticide-intensive maize field. When Khun Kul arrived, soil tests found arsenic and significant chemical residues across the whole 50-rai property.

She remediated the soil with locally abundant organic materials (leaf litter, rice straw, bamboo and rubber-tree leaves, dry compost and corn husks) and developed seven herbal fermentation formulas for soil improvement, pest repulsion and root nourishment. Soil is prepared in five steps: dig, break up, mulch, feed with dry and liquid fertilizer, then leave to ferment for a week before planting. The process also lends itself to science lessons on fungal communities.

Keeping every drop of rain

"In Nan, some 87.2 percent of the mountainous terrain consists of upstream forests that are crucial to the flow of the Chao Phraya," Kul explains.

The Khok Nong Na model manages water storage using natural-farming principles, a win-win in which both forests and farmers benefit. Three principles guide the site: retain water in the soil, capture 100% of rainfall, and build biodiversity, the central pillar of ChTN's work.

The paddies were dug into laterite mixed with gravel using traditional techniques: bunds raised with the excavated soil, irrigation channels along their base, vegetables on the bund tops, and fish raised in the fields with sawian, bundles of dry leaves and manure submerged to grow plankton as fish food. Results were visibly better than on neighboring plots, and students from Rajamangala University of Technology Lanna (RMUTL) Nan came to study the rice-growing process here.

The five-level forest

Starting from bare land, Mrs Kul planted trees across five vertical strata:

1. **Tall trees —** teak, makha, phayung
2. **Mid-level fruit trees —** longan, lychee and more than 30 other species
3. **Understory —** cocoa, coffee
4. **Low and creeping plants —** medicinal herbs, pumpkin, pepper
5. **Roots and tubers —** yams and other underground crops

Elsewhere on the site, an orchard due to be converted from longan to rubber was instead left to regenerate. Within a year of stopping herbicide use, 22 tree species had returned on their own from surviving roots, rhizomes and seeds, illustrating King Rama IX's idea of "planting forest without planting". Bird diversity has risen from about 20 species before Mrs Kul arrived to more than 100 today, in keeping with the center's principle: "nourish the soil so trees grow; let trees nourish people."





Activities and Training Content

Three training modules

Mrs Kul uses different tools for different groups in a three-pronged education plan:

1. **Field observation —** for visitors who want to see what happens on the ground; the aim is a change of mindset by learning through seeing and doing.
2. **Training course —** five days and four nights of living, eating and learning according to the Khok Nong Na model, networking with like-minded people.
3. **Volunteering —** course graduates stay at least two weeks to learn to create a Khok Nong Na suited to their own means, and to spread the approach to others.

Courses cover land design and site planning, compost making, household products such as multipurpose cleaning liquid, and traditional health practices.

Circular living and zero waste

Whether people come as students, volunteers or observers, Mrs Kul wants them to grasp the value of circular living. All the food served at the center is grown on site, and leftovers are composted into organic fertilizer. Catering is planned to create no waste: menus are simple dishes made from local ingredients such as fish from the pond and organic vegetables from the garden, packaging is made from natural materials such as banana leaves. Kitchens cook with bio-charcoal, lights run on solar cells and water is circulated by solar-powered pumps, costly to install but worth it for the environment in the long run.

Self-reliance, step by step

The center follows King Rama IX's nine-step ladder: first become self-sufficient in daily necessities (food, medicine and household products), then move on to surplus, sharing and finally production for sale. Khun Kul encourages visitors to aim for at least 25% self-reliance in everyday items such as soap, shampoo and dishwashing liquid. ChTN's own products include organic kaffir-lime-leaf shampoo, turmeric and centella soaps, soy sauce, a "green oil" herbal balm and activated charcoal from branches for cosmetics; during COVID-19 the center produced herbal medicine tablets.

A refuge in times of disaster

ChTN is designed to shelter 300 people for a month in the event of a natural disaster and to serve as a regional relief-distribution hub, with space for tents, food stocks and herbal medicines.

TIMELINE SUMMARY

2013
B.E. 2556
ChTN founded on a 50-rai plot; soil restoration begins.

2016
B.E. 2559
Training programs start generating income.

2018
B.E. 2561
Mrs Kul presents ChTN's water work at the UN on SDG 6.

2020–21
B.E. 2563–64
COVID-19: closed to visitors; 70,000 herbal tablets produced.

2022
B.E. 2565
Featured in Thailand's BCG book for the APEC summit.

2024
B.E. 2567
Bike For Earth hosted in June; Mrs Pui settles at ChTN

2025
B.E. 2568
Site renovated; Araya Camp and Forest Restoration trainings.

2028
B.E. 2571
Goal: ChTN 100% self-financing from its own organic products.



